Predictive Maintenance and Anomaly Detection in 6G-Integrated Smart Manufacturing Systems: An Isolation Forest Approach

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Abstract—**Smart manufacturing environments face substantial operational risk from unexpected machine failures, which are difficult to predict using traditional rule-based threshold monitoring. This paper presents a machine learning framework for predictive maintenance and anomaly detection in a 6G-integrated industrial facility operating 50 machines over a 69-day observation window from January to March 2025. Using 100,000 sensor records sampled at one-minute intervals, we engineer 17 process-derived features – including per-machine rolling deviation baselines, vibration-to-power instability ratios, error escalation trends, and network stress indices – and apply Isolation Forest anomaly detection to identify abnormal operational states without labeled failure data. The model detects 3,503 anomalies across 70,054 active-mode records, which are mapped to three maintenance risk levels: Low, Medium, and High. All 50 machines exhibit at least one high-risk event during the observation window, with Machine 18 recording the highest event frequency (27 events) and Machine 28 recording the maximum anomaly score of 1.000. A temporal spike in medium-risk events is identified on March 1, 2025, suggesting a system-wide operational stress event. Production speed and error rate are identified as the strongest discriminators of machine efficiency status. An interactive Streamlit dashboard operationalizes the detection framework for real-time monitoring. Results demonstrate that unsupervised anomaly detection on engineered sensor features provides actionable early warning signals ahead of efficiency degradation.**

Index Terms—**predictive maintenance, anomaly detection, Isolation Forest, smart manufacturing, 6G networks, sensor analytics, unsupervised learning**

I. Introduction

The transition from Industry 4.0 toward Industry 5.0 has reshaped the manufacturing floor into a densely instrumented environment in which temperature, vibration, power draw, and network telemetry are captured continuously from every machine on the line. This shift is driven in large part by advances in wireless communication infrastructure, and fifth- and sixth-generation (5G/6G) networks in particular are increasingly positioned as the connective layer that allows thousands of industrial sensors to stream data with sub-millisecond latency and near-deterministic reliability. In a 6G-integrated manufacturing facility, every machine effectively becomes a node in a real-time data fabric, and the resulting volume and velocity of sensor data offer an unprecedented opportunity to observe equipment health as it evolves rather than as it is periodically inspected.

Despite this abundance of data, most industrial monitoring systems in production today still rely on static, rule-based threshold alerts: a machine is flagged only when a single sensor reading crosses a predefined boundary, such as a temperature ceiling or a vibration limit. This approach is intuitive and computationally inexpensive, but it is structurally limited. Thresholds are typically calibrated to the population of machines as a whole rather than to the idiosyncratic operating envelope of an individual unit, so a machine that is drifting away from its own historical baseline may never breach a fixed threshold until failure is imminent. Moreover, threshold rules evaluate each sensor independently and are therefore blind to multivariate interactions, such as a simultaneous rise in vibration and network latency that, in combination, signals instability far earlier than either signal would in isolation.

This limitation carries real economic weight because unplanned machine failures, while comparatively rare events, are disproportionately costly. A single unscheduled stoppage can halt an entire production line, and the diagnostic and repair overhead associated with reactive maintenance routinely exceeds the cost of a scheduled intervention by a wide margin. Prior work in industrial reliability engineering has consistently observed that subtle deviations in operating parameters – small, cumulative departures from a machine's normal behavioral envelope – tend to precede catastrophic breakdowns by hours or days, well before any single sensor reading would trigger a conventional alarm. Capturing these precursors requires a monitoring approach that models what is normal for each machine individually and flags departures from that norm, rather than comparing raw readings against a fleet-wide static rule.

A further gap in current practice is conceptual rather than technical. Most manufacturing analytics platforms are oriented toward describing current efficiency – reporting whether a machine is presently operating at high, medium, or low output – rather than estimating the risk of future failure. Efficiency status and failure risk are related but distinct quantities, and a machine can appear efficient in its output metrics while simultaneously exhibiting sensor-level instability that foreshadows a near-term fault. Bridging this gap requires a framework that treats anomaly detection and efficiency monitoring as complementary but separate problems.

This paper addresses these limitations by proposing an unsupervised anomaly detection framework built on engineered, physically motivated features derived from raw sensor telemetry collected in a 6G-integrated smart manufacturing environment. Rather than relying on labeled failure events, which are rarely available in sufficient volume in real industrial settings, the framework uses the Isolation Forest algorithm to identify operational states that are statistically rare relative to a machine's own rolling behavioral baseline and relative to the fleet as a whole. The resulting anomaly scores are mapped onto interpretable risk tiers that can be directly consumed by maintenance planning teams, and the entire framework is operationalized through an interactive dashboard suitable for real-time deployment.

The remainder of this paper is organized as follows. Section II describes the dataset, its structure, and exploratory findings that motivate the feature engineering strategy. Section III details the methodology, including baseline behavior modeling, the seventeen engineered features, and the Isolation Forest configuration. Section IV presents the anomaly detection results, high-risk machine analysis, and temporal patterns observed across the 69-day window. Section V discusses the implications of these findings, including the network-stress-driven nature of anomalies in this environment. Section VI describes the Streamlit-based operational dashboard. Section VII concludes the paper and outlines directions for future work.

II. Dataset Description

The dataset used in this study, the Thales Group 6G-Integrated Smart Manufacturing dataset, comprises 100,000 sensor records collected from a facility operating 50 distinct machines, uniquely identified by `Machine_ID` values ranging from 1 to 50. Data collection follows a one-record-per-minute-per-machine sampling scheme over a continuous 69-day window spanning January 1, 2025 through March 10, 2025, consistent with uninterrupted business operation across the observation period. The raw dataset contains fourteen fields, summarized in Table I, and exhibits zero null values and zero duplicate records, indicating a clean and well-curated collection pipeline.

TABLE I
Dataset Fields and Descriptions

Field	Description
Date	Calendar date of the recorded observation
Time / Timestamp	Minute-level timestamp of the reading
Machine_ID	Unique machine identifier (1–50)
Operation_Mode	Active, Idle, or Maintenance state
Temperature_C	Machine temperature, 30–90 °C
Vibration_Hz	Vibration frequency, 0.1–5.0 Hz
Power_Consumption_kW	Power draw, 1.5–10.0 kW
Network_Latency_ms	6G network round-trip latency, 1–50 ms
Packet_Loss_%	Network packet loss, 0–5%
QC_Defect_Rate_%	QC-detected defect rate, 0–10%
Production_Speed	Throughput, 50–500 units/hr
Maintenance_Score	AI-derived health score, 0–1
Error_Rate_%	Operational error rate, 0–15%
Efficiency_Status	Target label: High, Medium, or Low

The `Operation_Mode` field partitions the dataset into three states: Active mode accounts for 70,054 records (70.1%), Idle mode accounts for 20,057 records (20.1%), and Maintenance mode accounts for 9,889 records (9.9%). The target label, `Efficiency_Status`, is highly imbalanced toward the Low class, which comprises 77,825 records (77.8%), followed by Medium with 19,189 records (19.2%) and High with only 2,986 records (3.0%). This imbalance reflects the operational reality that sustained high-efficiency states are comparatively rare relative to baseline and moderate throughput operation.

A. EDA Findings

Exploratory data analysis reveals a striking structural property of the sensor data: all nine continuous sensor features exhibit near-zero pairwise correlation, with every off-diagonal entry in the correlation matrix falling below 0.02 in absolute value. This near-total independence among raw sensor channels is consistent with a synthetically generated 6G simulation environment in which each sensor stream is sampled independently rather than governed by a shared underlying physical process, and it has direct methodological consequences: single-sensor thresholding cannot exploit cross-sensor correlation structure, and feature engineering must instead construct informative combinations explicitly rather than relying on the raw features to co-vary naturally.

Among the fourteen raw fields, only two exhibit meaningful separation across the three efficiency classes. Production speed rises monotonically with efficiency, averaging 254.9 units/hr for Low-efficiency records, 334.1 units/hr for Medium, and 450.8 units/hr for High. Error rate falls just as sharply in the opposite direction, averaging 8.930% for Low, 2.733% for Medium, and 1.007% for High. The remaining seven sensor channels – temperature, vibration, power consumption, network latency, packet loss, defect rate, and the predictive maintenance score – show near-identical distributions across all three efficiency classes, offering little discriminative signal for efficiency classification on their own. This asymmetry, in which only two of nine sensors carry class-discriminative information, further motivates the construction of derived and temporal features described in Section III, since the raw sensor channels alone are insufficient to characterize abnormal machine states.

Because Idle and Maintenance operation modes represent qualitatively different operating regimes with their own baseline sensor signatures, including them alongside Active-mode records would confound the anomaly detection process: a machine legitimately powering down for scheduled maintenance would appear anomalous relative to Active-mode statistics despite representing entirely normal behavior. Accordingly, all anomaly detection modeling described in this paper is applied exclusively to the 70,054 Active-mode records.

Figure 1 shows the marginal distribution of each of

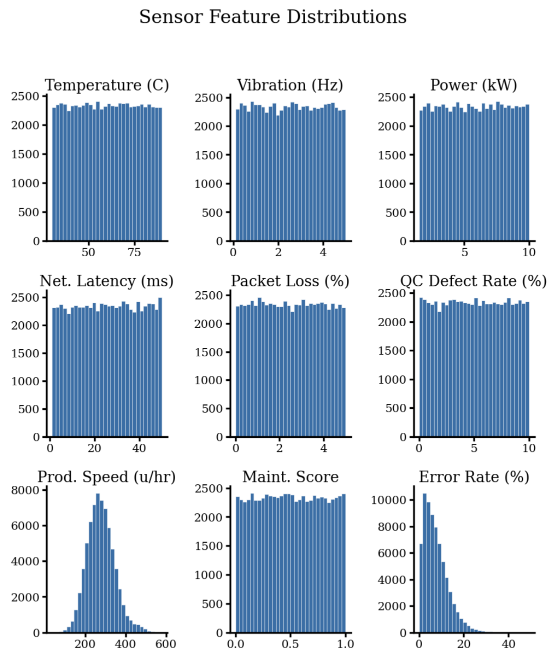

Fig. 1. Marginal distributions of the nine raw sensor channels across the 70,054 Active-mode records.

the nine sensor channels across the Active-mode dataset. Seven of the nine channels are approximately uniform across their specified operating ranges, consistent with the near-zero inter-feature correlation reported above, while Production_Speed_units_per_hr and Error_Rate_% depart from uniformity and instead show the multi-modal and right-skewed shapes, respectively, that reflect the underlying mixture of efficiency classes described in this subsection.

III. Methodology

The overall methodology proceeds in four stages: establishing a per-machine behavioral baseline, engineering a seventeen-dimensional feature representation from that baseline and the raw sensor channels, fitting an Isolation Forest model to the resulting feature space, and mapping the resulting anomaly scores onto interpretable maintenance risk tiers.

A. Baseline Behavior Modeling

Because the nine raw sensor channels are drawn from fixed operating ranges with negligible inter-feature correlation, absolute sensor values carry limited information about whether a given machine is behaving normally for itself. A machine that consistently operates near the upper end of the temperature range is not necessarily unhealthy; what matters is whether its current reading deviates from its own recent operating history. To capture this, a rolling baseline is computed independently for each machine using a 60-minute trailing window. For each of the nine sensor features, the rolling mean $\mu_{i,t}$ and rolling standard deviation $\sigma_{i,t}$ are computed at each timestamp t for machine i, with a minimum warm-up period of ten observations required before a baseline is considered valid. The instantaneous deviation of machine i from its own baseline is then expressed as a z-score,

$$z_{i,t} = \frac{x_{i,t} - \mu_{i,t}}{\sigma_{i,t}}, \tag{1}$$

where $x_{i,t}$ denotes the raw sensor reading for machine i at time t. This formulation reframes every sensor channel from an absolute measurement into a self-referential deviation signal, which is the foundation for the first group of engineered features described below.

B. Feature Engineering

A total of seventeen features are engineered from the raw fourteen-column dataset, organized into three functional groups.

Group 1 – Sensor Deviation Features (9 features). For each of the nine continuous sensor channels – Temperature_C, Vibration_Hz, Power_Consumption_kW, Network_Latency_ms, Packet_Loss_%, Quality_Control_Defect_Rate_%, Production_Speed_units_per_hr, Predictive_Maintenance_Score, and Error_Rate_% – the 60-minute rolling z-score deviation defined in Eq. (1) is computed per machine, yielding nine deviation features that express how far each channel currently sits from that specific machine's recent norm.

Group 2 – Derived Ratio Features (6 features). Because the raw sensors are nearly uncorrelated, six physically motivated combination features are constructed to capture multivariate instability that no single channel would reveal on its own. The vibration-power ratio identifies mechanical instability disproportionate to power draw,

$$R_{\text{vib-pow}} = \frac{\text{Vibration\_Hz}}{\text{Power\_Consumption\_kW} + 0.001}, \tag{2}$$

where the additive constant prevents division by zero. Error and defect signals are combined multiplicatively to amplify joint quality degradation,

$$C_{\text{err-def}} = \text{Error\_Rate\_\%} \times \text{Quality\_Control\_Defect\_Rate\_\%}. \tag{3}$$

Network health is captured through a network stress index that penalizes latency more heavily as packet loss increases,

$$S_{\text{net}} = \text{Network\_Latency\_ms} \times (1 + \text{Packet\_Loss\_\%}). \tag{4}$$

The AI-derived predictive maintenance score is inverted so that higher values consistently indicate greater risk across all engineered features,

$$M_{\text{inv}} = 1 - \text{Predictive\_Maintenance\_Score}. \tag{5}$$

Thermal and mechanical stress are combined into a single interaction term,

$$T_{\text{stress}} = \frac{\text{Temperature\_C} \times \text{Vibration\_Hz}}{100}, \tag{6}$$

and production throughput is normalized against its maximum observed range to yield a bounded efficiency ratio,

$$E_{\text{speed}} = \frac{\text{Production\_Speed\_units\_per\_hr}}{500}. \tag{7}$$

Group 3 – Temporal Trend Features (2 features). To capture whether a machine's condition is actively deteriorating rather than merely deviating momentarily, two short-versus-long window trend features are computed by differencing a 30-minute rolling mean against a 120-minute rolling mean. The error escalation trend is defined as

$$\Delta_{\text{err}} = \overline{\text{Error\_Rate}}_{30\text{min}} - \overline{\text{Error\_Rate}}_{120\text{min}}, \tag{8}$$

and the maintenance score decay is defined analogously using the predictive maintenance score,

$$\Delta_{\text{PMS}} = \overline{\text{PMS}}_{30\text{min}} - \overline{\text{PMS}}_{120\text{min}}. \tag{9}$$

Positive values of Δ_{err} indicate that recent error behavior exceeds the longer-run average, signaling an escalating trend rather than a one-off spike, while negative values of Δ_{PMS} indicate that the machine's predicted health is declining faster in the near term than its longer-run trajectory would suggest.

C. Isolation Forest

Anomaly detection is performed using the Isolation Forest algorithm of Liu et al. [1], an ensemble method that isolates observations by recursively partitioning the feature space with randomly selected split features and split values. Anomalous points, being few and different, tend to require fewer partitions to isolate and therefore receive shorter average path lengths across the ensemble of trees, which the algorithm converts into an anomaly score.

The model is fitted exclusively on the 70,054 Active-mode records using all seventeen engineered features, which are standardized with a StandardScaler prior to fitting so that features on different numeric scales, such as the network stress index and the bounded z-score deviations, contribute comparably to the partitioning process. The ensemble is configured with $n_{\text{estimators}} = 200$ trees, a contamination parameter of 0.05 (reflecting a prior expectation that approximately five percent of Active-mode observations represent anomalous operating states), and a fixed random seed of 42 for reproducibility.

The raw isolation score returned by the model, `score_samples`, is transformed into an anomaly score bounded on $[0, 1]$ via min-max normalization and inversion,

$$a_{i,t} = 1 - \frac{s_{i,t} - s_{\min}}{s_{\max} - s_{\min}}, \tag{10}$$

where $s_{i,t}$ is the raw score-samples output for machine i at time t, and $s_{\min}$, $s_{\max}$ are the minimum and maximum raw scores observed across the Active-mode dataset. Under this transformation, higher values of $a_{i,t}$ correspond to greater anomalousness, which is more intuitive for downstream risk communication than the raw isolation score.

The continuous anomaly score is then discretized into three operational risk tiers using fixed thresholds,

$$\text{Risk}(a_{i,t}) = \begin{cases} \text{High}, & a_{i,t} \geq 0.70 \\ \text{Medium}, & 0.40 \leq a_{i,t} < 0.70 \\ \text{Low}, & a_{i,t} < 0.40 \end{cases} \tag{11}$$

These thresholds were selected to yield operationally interpretable tiers rather than to optimize any single statistical criterion, prioritizing a small, actionable High-risk tier that maintenance teams can realistically inspect within a normal operating cycle.

D. Feature Importance

Feature importance is assessed by ranking the seventeen engineered features according to their mean absolute value across all Active-mode records, which reflects each feature's typical magnitude of contribution to the isolation partitioning process. The network stress index dominates this ranking with a mean absolute value of 89.16, more than double the next-ranked feature, the combined error-defect term, at 37.53. The temperature-vibration stress interaction follows at 1.53, and the remaining sensor deviation features cluster tightly around 0.85, led by the quality control defect rate deviation at 0.855 and the production speed deviation at 0.854. This ordering indicates that the two network-derived engineered features – which have no counterpart among the raw sensor channels – carry substantially more weight in the anomaly detection process than any individual per-sensor deviation term.

IV. Results

A. Anomaly Detection Results

Applying the Isolation Forest model to the 70,054 Active-mode records produces the summary statistics reported in Table II. The model labels 3,503 records as anomalous, corresponding to 5.0% of Active-mode observations, consistent with the specified contamination parameter, while the remaining 66,551 records (95.0%) are labeled normal. When anomaly scores are discretized into risk tiers, 831 records (1.19%) fall into the High-risk category, 27,390 records (39.10%) fall into Medium risk, and 41,833 records (59.72%) fall into Low risk. The mean anomaly score across all Active-mode records is 0.3763, and the maximum observed score is 1.0000, attained by Machine 28. Critically, all 50 machines in the facility exhibit at least one High-risk event during the 69-day observation window, indicating that elevated risk is a fleet-wide phenomenon rather than one confined to a small subset of underperforming units.

Figure 2 presents the distribution of anomaly scores across the full Active-mode population alongside the resulting counts within each risk tier. The score distribution is right-skewed, with the bulk of observations concentrated below the Medium-risk threshold and a long tail of increasingly rare, high-score observations extending toward the maximum. This shape is consistent with the underlying assumption of Isolation Forest,

TABLE II
MODEL RESULTS SUMMARY

Metric	Value
Total Active Records	70,054
Anomalies Detected (-1 label)	3,503 (5.0%)
Normal Records (1 label)	66,551 (95.0%)
High Risk Events	831 (1.19%)
Medium Risk Events	27,390 (39.10%)
Low Risk Events	41,833 (59.72%)
Average Anomaly Score	0.3763
Peak Anomaly Score	1.0000 (Machine 28)
Machines with $\geq$1 High-Risk Event	50 of 50 (100%)

TABLE III
TOP 10 HIGH-RISK MACHINES

Machine	Events	Avg. Score	Peak Score
Machine 18	27	0.748	0.857
Machine 32	27	0.767	0.889
Machine 3	25	0.756	0.859
Machine 33	22	0.745	0.816
Machine 24	22	0.750	0.895
Machine 44	21	0.762	0.893
Machine 14	21	0.738	0.815
Machine 20	21	0.755	0.881
Machine 7	20	0.751	0.928
Machine 47	19	0.750	0.953

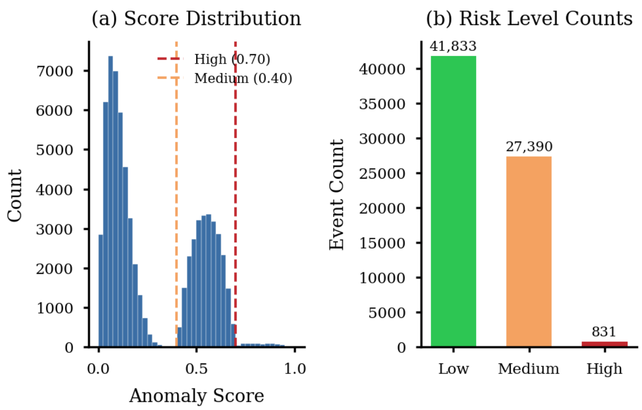

Fig. 2. Distribution of anomaly scores across all Active-mode records (a) and resulting counts within each risk tier (b). Dashed lines in (a) mark the High-risk (0.70) and Medium-risk (0.40) thresholds.

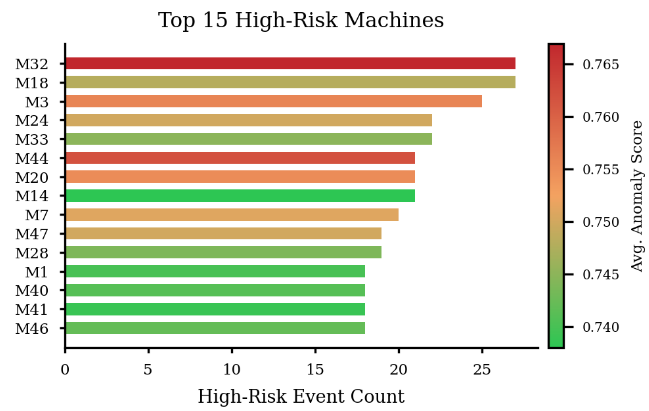

Fig. 3. Top 15 machines ranked by High-risk event count, color-coded by average anomaly score.

in which the majority of the population is treated as the operating norm and only a small fraction of genuinely distinctive observations receive elevated scores.

B. High-Risk Machine Analysis

Table III lists the ten machines with the highest count of High-risk events during the observation window. Machines 18 and 32 tie for the highest event frequency at 27 High-risk events each, followed closely by Machine 3 with 25 events. Average anomaly scores among these top machines remain tightly clustered in the range of approximately 0.74 to 0.77, indicating that the machines experiencing the most frequent High-risk events do not necessarily experience the most extreme individual anomaly scores; Machine 47, for instance, ranks tenth by event count but records the highest peak score of 0.953 among the top ten.

Figure 3 extends this ranking to the top fifteen machines and color-codes each bar by average anomaly score, revealing that event frequency and average severity are only loosely coupled across the fleet: several machines with comparatively fewer High-risk events, such as Machine 41 and Machine 40, nonetheless post average scores comparable to the most frequently flagged units.

C. Temporal Analysis

The daily distribution of risk events remains largely stable across January and February, with day-to-day fluctuation consistent with ordinary operating variability. A pronounced exception occurs on March 1, 2025, when the count of Medium-risk events rises to approximately twice the typical daily count, a deviation large enough to stand out clearly against the otherwise stable temporal baseline shown in Fig. 4. Because this spike manifests simultaneously across many machines rather than being confined to a single unit, it is more consistent with a system-wide operational stress event or an environmental or network-level perturbation than with the independent degradation of any one machine.

Examining the intraday distribution of High-risk events reveals no concentration within any particular shift or hour; High-risk events are distributed approximately uniformly across all 24 hours of the day, suggesting that elevated risk is not attributable to shift-specific operating practices, staffing patterns, or time-of-day environmental conditions. On a weekly basis, between six and nine distinct machines are affected by at least one High-risk event in any given week throughout the observation window, indicating a persistent and recurring, rather than episodic, pattern of risk distribution across the fleet.

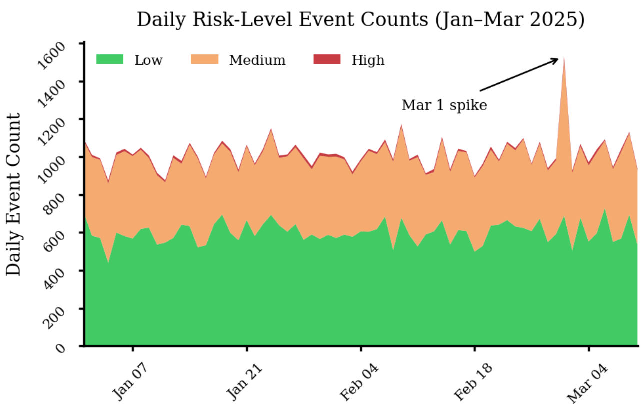

Fig. 4. Stacked daily counts of Low-, Medium-, and High-risk events from January 1 to March 10, 2025, with the March 1 Medium-risk spike annotated.

D. Sensor Behavior During High-Risk Events

Comparing sensor readings recorded during High-risk windows against the same machine's normal operating baseline reveals systematic elevation in several channels. Temperature_C and Network_Latency_ms are both measurably higher during High-risk periods relative to each machine's typical operating range, and Error_Rate_% and Vibration_Hz show even more pronounced elevation, consistent with their prominence in the feature importance ranking reported in Section III-D. Across the fleet, the network stress index and the combined error-defect term are the two engineered features most responsible for elevating the anomaly score during High-risk events, reinforcing the conclusion that network-layer instability and compounding quality degradation, rather than any single raw sensor reading, are the principal drivers of anomalous operational states in this environment.

V. Discussion

The finding that all 50 machines in the facility recorded at least one High-risk event over the 69-day window is, on its own, an important departure from the conventional predictive maintenance narrative in which failure risk is expected to concentrate in a small number of chronically underperforming units. Instead, the near-universal incidence of High-risk events suggests that the 6G-integrated environment itself may periodically introduce fleet-wide stressors – for instance, transient network congestion or latency spikes affecting many machines simultaneously – rather than risk arising purely from isolated, machine-specific mechanical degradation. This reframes the maintenance problem: rather than asking which machines are unreliable, the more productive question in this environment may be which conditions, external to any single machine, precipitate correlated risk across the fleet.

The March 1, 2025 spike in Medium-risk events is a particularly instructive finding in this regard. A genuine anomaly attributable to independent machine-level degradation would be expected to manifest gradually and asynchronously across different units as each machine's condition deteriorates on its own timeline. The observed pattern instead shows risk events clustering sharply on a single calendar day across many machines simultaneously, a signature far more consistent with a shared, correlated cause – such as a network-wide disturbance, a facility-level environmental event, or a scheduling anomaly – than with independent mechanical failure. This distinction has direct operational implications: a correlated event of this kind warrants investigation of shared infrastructure, such as the 6G network backbone or facility power supply, rather than dispatching maintenance personnel to inspect individual machines in isolation.

The dominance of the network stress index in the feature importance ranking, at more than double the magnitude of the next most important feature, is itself a novel finding specific to the 6G-integrated manufacturing context. It indicates that network latency and packet loss, rather than the traditionally emphasized mechanical sensors such as vibration and temperature, are the primary drivers of anomalous behavior as detected by the model in this environment. This has practical significance for facilities adopting 6G connectivity for industrial telemetry: network quality of service becomes not merely an infrastructure concern but a leading indicator of operational anomaly, and monitoring systems designed for such facilities should treat network-layer metrics as first-class predictive features rather than auxiliary diagnostics.

The near-total independence of the nine raw sensor channels, with all pairwise correlations below 0.02, further underscores the value of the engineered combination features described in Section III-B. Because no pair of raw sensors co-varies meaningfully, a monitoring system relying solely on single-sensor thresholds would be structurally unable to detect the kind of joint instability that engineered features such as the vibration-power ratio and the combined error-defect term are specifically designed to surface. This is a broader methodological lesson applicable beyond this dataset: in sensor environments exhibiting low inter-feature correlation, explicit feature engineering that constructs meaningful combinations is not merely beneficial but necessary for multivariate anomaly detection to succeed.

The fixed risk classification thresholds of 0.70 and 0.40 provide maintenance teams with three operationally distinct tiers rather than a single undifferentiated alarm. High-risk events, comprising just 1.19% of Active-mode records, define a tractable inspection queue that can realistically be reviewed within a normal maintenance cycle. Medium-risk events, at 39.10% of records, are better suited to passive monitoring and trend tracking rather than immediate intervention, while Low-risk records require no action. This tiered structure allows the detection framework to be integrated into existing maintenance workflows without requiring a wholesale change in operational practice.

A central limitation of the present approach warrants explicit acknowledgment. Isolation Forest assigns anomaly status on the basis of statistical rarity within the observed feature space, not on the basis of any verified downstream failure outcome, since no labeled failure or breakdown data was available for this facility. An anomalous operating state, as detected here, is therefore best interpreted as an early warning signal rather than a confirmed precursor to failure. Establishing the true predictive validity of these anomaly scores – that is, quantifying how often a High-risk classification is followed by an actual maintenance event or breakdown within a defined horizon – requires validation against real maintenance logs, and this remains an important direction for future work.

VI. Dashboard Implementation

To operationalize the detection framework for practical use by maintenance planning teams, the modeling pipeline is deployed behind an interactive dashboard built with Streamlit and Plotly, organized into four functional tabs.

The Maintenance Overview tab provides a fleet-wide summary through five key performance indicator cards reporting total records processed, High- and Medium-risk event counts, the average anomaly score across the current filtered view, and the number of machines currently at elevated risk. Below the indicators, a risk distribution pie chart summarizes the proportion of records in each risk tier, a horizontal bar chart ranks the top fifteen machines by average anomaly score, a histogram of anomaly scores displays the High- and Medium-risk threshold markers for context, and a supplementary panel breaks down risk incidence by operation mode.

The Machine Anomaly Dashboard tab supports per-machine investigation through a machine selector control. Upon selecting a machine, the dashboard renders an anomaly score timeline across the observation window with High-risk events explicitly marked, a radar chart summarizing the selected machine's z-score deviations across all nine sensor channels, and a comparative bar chart contrasting sensor readings recorded during High-risk events against the machine's normal operating baseline.

The Maintenance Alert Panel tab is designed for immediate operational triage. It presents a warning banner summarizing current fleet risk status, a priority inspection queue table that labels machines as Critical or High based on recent anomaly activity, a table of the fifty most recent High-risk events with associated timestamps and scores, and a downtime prevention index chart that estimates the potential operational value of acting on the current alert queue.

The Historical Risk Analysis tab supports longer-horizon pattern review through a stacked area chart of daily risk-tier distribution across the full observation window, an escalation timeline for any selected machine showing how its anomaly score has trended over time, a stacked bar chart of hourly risk distribution to surface any time-of-day patterns, and a dual-axis weekly chart that jointly tracks the count of High-risk events and the number of distinct machines affected each week.

Across all four tabs, a persistent sidebar provides global filtering controls, including a multi-select machine filter, a risk threshold slider adjustable between 0.5 and 0.95 to allow maintenance teams to tune sensitivity, an operation mode filter, and a date range picker for restricting analysis to a specific time window. This filtering layer allows the same underlying model output to serve both broad fleet-level monitoring and narrow, machine-specific diagnostic review within a single interface.

VII. Conclusion

This paper presented an unsupervised anomaly detection framework for predictive maintenance in a 6G-integrated smart manufacturing environment, built on seventeen engineered features spanning per-machine sensor deviations, physically motivated ratio and interaction terms, and short-versus-long window temporal trends. Applied to 70,054 Active-mode sensor records via Isolation Forest, the framework identified 3,503 anomalous observations, which were mapped onto three interpretable risk tiers to support maintenance prioritization. All 50 machines in the facility exhibited at least one High-risk event, and the network stress index emerged as the dominant driver of anomaly scores by a wide margin, indicating that network-layer telemetry – latency and packet loss specific to the 6G communication fabric – carries more predictive weight for anomalous operation in this environment than traditional mechanical sensors such as vibration or temperature. A pronounced, fleet-wide spike in Medium-risk events on March 1, 2025 further suggested that risk in this environment can arise from correlated, system-wide perturbations rather than solely from independent machine-level degradation, a pattern with direct implications for how maintenance investigations should be prioritized and scoped.

Beyond the modeling results, the Streamlit-based dashboard described in Section VI demonstrates that the detection framework can be translated into a practical operational tool, giving maintenance teams fleet-level situational awareness, per-machine diagnostic depth, and an actionable inspection queue within a single interface. Together, these results indicate that unsupervised anomaly detection on carefully engineered sensor features can provide meaningful early warning ahead of efficiency degradation, even in the absence of labeled failure data.

Future work should prioritize validating the anomaly scores produced by this framework against real maintenance and failure logs, in order to quantify how reliably a High-risk classification precedes a genuine mechanical or operational fault and to calibrate the risk thresholds accordingly. A deep learning autoencoder variant of the detection pipeline, in which reconstruction error serves as an alternative anomaly signal, represents a promising avenue for comparison against the Isolation Forest baseline established here. Finally, integrating the framework with a live 6G data stream, rather than the static historical dataset used in this study, would allow the detection and dashboard system described here to operate as a genuine real-time monitoring solution within an operational manufacturing facility.

Acknowledgment

This work was completed as part of a project-based machine learning internship with Unified Mentor Pvt. Ltd.

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